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Noninvasive continuous blood pressure estimation with fewer parameters based on RA-ReliefF feature selection and MPGA-BPN models

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ABSTRACT

Photoplethysmography (PPG) and electrocardiogram (ECG) are closely related to the cardiovascular system and have substantial waveform parameters associated with blood pressure (BP), so extracting their waveform parameters to estimate BP has been widely studied. However, the more parameters, the more complex the model is, and the more factors affect the estimation results, which further affects the application of the BP estimation model to related equipment. Greatly reducing the number of parameters without reducing the accuracy of BP prediction, so as to improve the availability, applicability and robustness of BP estimation model, is the key for the real application of this method. In this study, we used relativity analysis (RA) to fill the gap of ReliefF algorithm, so the same number of parameter dimensions selected by ReliefF contain more signal information. Furthermore, a BP prediction model based on multi-population genetic algorithm (MPGA) and back propagation network (BPN) was constructed. Using waveform parameters of PPG and ECG signals with a duration of 20 s and the measured BP as a set of data, we extracted 522 and 23,608 sets of data from 25 healthy subjects and the UCI_BP dataset respectively. Three, five, seven and all parameters were selected by ReliefF for training in MPGA-BPN model and the results were compared. The comparison shows that five-parameters dataset can performs as all- parameters in both healthy individuals and UCI_BP, which confirms the validity of the feature selection in this study and lays the roots for BP estimation with fewer parameters and high accuracy.

1. Introduction

According to statistics released by the World Health Organization, hypertension, the leading cause of premature death worldwide, is a serious factor that increases the risk of heart, brain, kidney and other diseases[1]. Unfortunately, due to the global population aging, the prevalence of hypertension has been increasing and has become a pressing global health problem[2,3]. Reliable and timely blood pressure (BP) measurement is essential for the prevention of hypertension and hypertension-related diseases. As BP regulation is a complex physiological mechanism, its fluctuation is influenced by physical conditions and emotions. Therefore, non-continuous BP measurement is not good at helping doctors to identify high-risk individuals, avoid “masked” or “white-coat” hypertension. Moreover, hypertension is a chronic disease requiring long-term care, so continuous BP monitoring can help doctors and patients to monitor the effectiveness of BP control.

The gold standard for continuous BP measurement is invasive catheterization. The catheter needs to be inserted into the heart or blood vessels, which can easily lead to a series of complications such as infection and bleeding, and this method is often used in ICU. Therefore, methods for noninvasive continuous BP (NCBP) measurement are being widely pursued for routine continuous BP monitoring. Among them, the PPG based method is regarded as a promising technology. PPG is generated by heartbeat and formed by blood traveling outward along the arteries. When the flowing blood squeezes against the vessel wall, the vessel deforms and vibrates, that leads to waveform changes related to BP changes. Therefore, PPG contains extremely rich cardiovascular information.

Volume clamping is a NCBP measurement method based on PPG[4]. Its principle is to place a cuff embedded with a PPG sensor around the finger. The PPG sensor measures fluctuations in blood volume, while the manometer records the cuff pressure. The cuff needs to provide pressure to keep the vessel unloaded during the measurement, which means that the blood vessel is constantly being compressed, and prolonged measurements can cause significant discomfort to the subject. Therefore, in recent years, many researchers have been working on the cuffless BP measurement with PPG.

There are two methods based on PPG for cuffless BP measurement:

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pulse transit time (PTT) and waveform analysis[5–7].

PTT is defined as the time interval between the waveform recorded by two sensors (PPG&PPG/ECG&PPG). It relates BP via Moens-Korteweg (MK)[8] and Huges equations[9].

$$\text{MK Equation}: \frac{D}{PTT} = \sqrt{\frac{Eh_0}{2\rho R_0}} \tag{1}$$

$$\text{Hughes Equation}: E = E_0 \exp(\xi P) \tag{2}$$

Where D is the distance between sensors. E, h_0 and R_0 are the elastic modulus, thickness and radius of blood vessel when the BP is P. ρ is the blood density. E_0 is the elastic modulus at zero BP and ξ is the material coefficient of the artery. Combining equation (1) and (2), we can get equation (3).

$$P = \frac{1}{\xi}\left[2\ln\left(\frac{D}{PTT}\right) + \ln\left(\frac{2\rho R_0}{E_0 h_0}\right)\right] \tag{3}$$

According to Equation (3), there is a certain mathematical relationship between PTT and BP. Therefore, many researches use PTT as input for regression models to estimate BP[10–12]. However, equation (1) involves two limited assumptions: i) the arterial wall is thin, so it can be modeled as a thin shell. ii) the thickness and radius of artery remain constant as BP changes. Equation (2) is empirical and has no theoretical basis[13]. Moreover, BP fluctuates with mood, physiological activities and etc. Therefore, it is not complete to rely only on PTT to fit the inherent fluctuation of BP, which will affect the prediction accuracy and reliability of model.

Considering the rich information in PPG and the complex physiological mechanism of BP, researchers tend to introduce more waveform features and more complex algorithms to estimate NCBP, namely waveform analysis, which is the main way of BP estimation at present.

Waveform analysis is studied in two steps, extracting features and constructing models. And the present models can be divided into two categories according to the temporal dependencies: point-to-point and sequence-to-point BP estimation models. Point-to-point models are the translation of a single point input of signal to point output. It means that the output at this moment has nothing to do with the input at other moments. Because the algorithm is timing independent, the computation is smaller than time series model. Common point-to-point BP estimation models include linear and nonlinear regression models[14–16], support vector machine models[17], random forests[17–19], and all models consisting of feedforward neural networks (BPN, CNN, etc.) [20,21]. Sequence-to-point models are the translation of time-series input to point output. The output BP at a given moment of a sequence-to-point model is determined by a combination of the input at that moment and previous input. In term of the nature of BP, the study of prediction of BP is essentially a study of prediction of time-series data. Therefore, time-series related models are gradually becoming a hot topic of current NCBP research. Among them, feedback neural network is the focus. Because the hidden layers of feedback neural networks are connected to each other and the inputs of the hidden layers include the outputs of the input layer and the outputs of the hidden layers at the previous moment, it provides a good basis for processing time-series data. The most common feedback neural networks for BP estimation are Recurrent Neural Network (RNN) and its variants Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU)[22–25].

Regardless of which of the above models is used for BP estimation, the form of model inputs has two ways: one is to extract features manually from PPG with the number of features between 10 and 20 [26,27], sometimes with fewer features, as few as 7[28], sometimes is more than 30[29]; the other uses deep learning network to automatically extract features[30]. The number of features is related to the number of network nodes, which may have more than 50 features.

For features extracted manually, the accuracy of extraction determines the accuracy of BP estimation. However, the accuracy of features extraction has always been a challenge due to the differences between PPGs in different populations[31,32], such as the difficulty of detecting dicrotic notch in hypertensive patients. The more features, the more uncertainty and the greater error rate will be in feature extraction, which affects the stability and accuracy of models. The number of features extracted automatically is large, and the features are directly used as the input of the BP estimation model without processing redundancy, which seriously limits the training efficiency of the model, increases the complexity of the model, and further affects the integration of the model on portable devices. Although more features mean that more information is used for BP estimation and the accuracy of BP estimation will be higher, it also means more redundancy, more feature points need to be located during feature extraction. Namely, there are more factors affecting the BP estimation in the whole process. Therefore, no matter how to extract features, the number of parameters needs to be further reduced.

In our previous study[27], 14 features were selected from the 24 features by the mean impact value (MIV) combined with the BPN, which reduces the model complexity in BP estimation. However, MIV-BPN requires several model simulations, so the time complexity of the feature selection algorithm is large, and the dimension of the features needs to be further reduced.

In order to reducing the time complexity of the feature selection algorithm, and the feature dimension, this study will use relativity analysis (RA) to eliminate highly correlated parameters, then further select parameters based on ReliefF algorithm with lower time complexity. Subsequently, BP model based on MPGA-BPN is built. In general, we hope to achieve the same or even higher accuracy of BP estimation with fewer parameters, which can not only increase the stability of the system, but also further simplify the BP estimation model and help to speed up the process of productization of NCBP methods.

Table 1
Feature definitions.

	Features	Definitions
Time indices	t_{up}	Time span from onset to peak
	t_{bf}	Time span from onset to dicrotic notch
	t_{ae}	Time span from the max slop of ascending branch to the min slop of descending branch
	t_{down}	Time span from peak to valley
	dt_{down}	
	t_{fb}	Time span from dicrotic notch to offset
	t_{bb}	Time span from onset to offset
	PTT_b	Pulse transit time (PTT): from R-wave of ECG to PPG points b, a, c, respectively.
	PTT_a	
	PTT_c	
	tw1	Time width at 1/6, 2/6, 3/6, 4/6, 5/6 of pulse amplitude of the PPG waveform, respectively.
	tw2	
	tw3	
	tw4	
	tw5	
Slope indices	S_{cf}	Slope from maximum peak to dicrotic notch
	S_{bc}	Slope from onset to maximum peak (Ascending slope)
	dS_{bc}	
	S_{cb}	Slop from maximum peak to valley (Descending slope)
	dS_{cb}	
Area indices	A1	Area under the PPG curve between selected points (onset, max slope, maximum peak, and offset), as shown in Fig. 1
	A2	
	A3	
	A4	
	A_{bf}	Area of systole
	A_{fb}	Area of diastole
	dAA	Ascending area
	dDA	Descending area
	K	$K = (A_m\text{-}I_c)/(I_c)$, A_m is the mean pulse wave area of the cycle. I_c is the intensity of c points
Intensity indices	I_{ar}	Ratio of selected points (a, e, f, g) intensity to peak intensity, respectively.
	I_{er}	
	I_{fr}	
	dPIR	Ratio of maximum peak intensity to onset intensity
others	HR	Heart rate

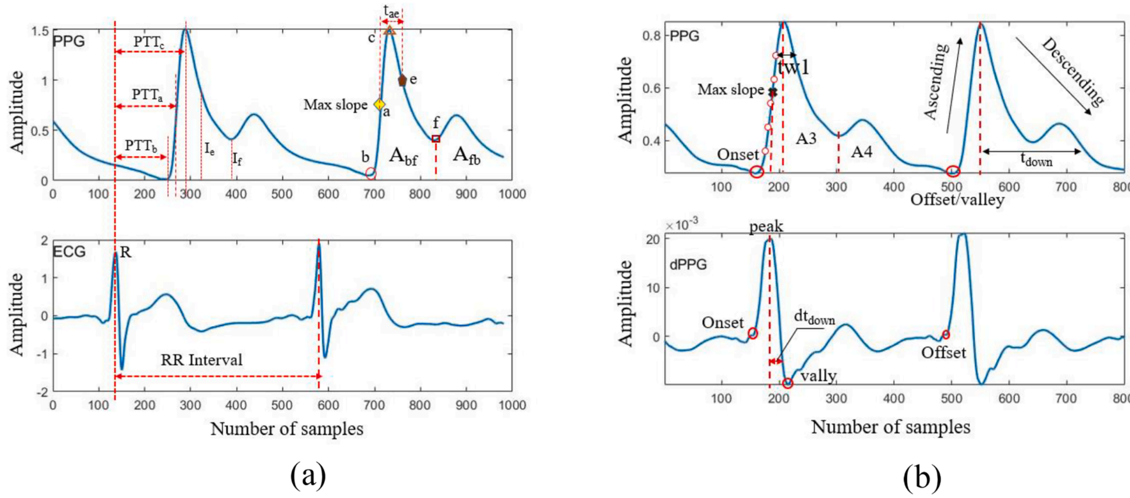

Fig. 1. Some features of signals. (a) PPG and ECG (b) PPG and its first derivate (dPPG).

2. Methodology

2.1. Data preprocessing and feature extraction

ECG and PPG are instable, because they are so weak that are susceptible to physiological activities, electromagnetic environment, and random factors.

In order to construct an accurate BP estimation model, the data should be accurate and reliable. Preprocessing is necessary for extracting features accurately and efficiently. We denoised the signals by wavelet thresholding algorithm in MATLAB. The specific process was as follows: i) selecting a wavelet function to decompose the signal; ii) performing threshold processing on the wavelet coefficients; iii) reconstructing the denoised signal with the processed wavelet coefficients.

According to the investigation on physiological mechanisms of PPG features for BP[27,33], 34 features of ECG and PPG were extracted in this study, as shown in Table 1. For extracting these features, some feature points were needed to be located. They were R waves in ECGs, b, a, c, e and f points in PPGs (they are onset, maximum ascending slope, peak, minimum descending slope and dicrotic notch points, respectively). The schematic diagram of feature points and some features is shown in Fig. 1.

2.2. Feature screening

2.2.1. Relativity analysis

When the dimension of features is too large, dimensions will be related to each other, which will affect the efficiency of model training. Also, too many features will complicate the model and require more computing resources, which is not conducive to the integration of the model on portable wearable devices. Furthermore, more parameters will bring more influencing factors, leading to the decreases of model stability and modeling accuracy. Therefore, we need to further select and analyze the 34 features mentioned above.

For the problem of interrelated features, Pearson correlation analysis is applied[34]. The Pearson coefficient between two variables can be calculated by the following equation:

$$\rho_{X,Y} = \frac{E[(X-\mu_X)(Y-\mu_Y)]}{\sigma_X\sigma_Y} \tag{4}$$

where $\rho_{X,Y}$ is the Pearson correlation coefficient between variable X and variable Y. $E[(X-\mu_X)(Y-\mu_Y)]$ is the covariance. σ_X, σ_Y are the standard deviation of X and Y.

If the two variables are perfectly linearly related, the absolute value of the Pearson coefficient is 1. Conversely, if the Pearson coefficient is equal to 0 or close to 0, the two variables are weakly or uncorrelated. In this study, Pearson correlation analysis was performed on the 34 features extracted, and the strong related features were removed according Pearson coefficients.

2.2.2. ReliefF

In addition to RA, ReliefF algorithm is used in further feature screening, that can assign different weights to features. Very important features for BP estimation can be determined by the weights, and the redundancy of features and the complexity of the model can be reduced. ReliefF combines the ideas of original Relief and K-nearest Neighbor algorithm[35]. The original Relief algorithm is a filter method. Its idea is as follows:

1. Constructing a vector statistic to measure the importance of features. Each component of the vector represents the weight of each feature, and the initial weight is set to 0;
2. Selecting an instance X from training set randomly;
3. Finding X's nearest hit (H) from the same class and X's nearest miss (M) from the other class;
4. If the distance of i-th feature between instance X and H is less than the distance between X and M, we can think that the i-th feature is positive for classification. Then we can increase the i-th weight. Conversely, if i-th feature is negative for classification, we can decrease its weight. the weight of i-th feature is calculated by equation (5);
5. Repeating step 4th m times to obtain the average value of each weight;
6. Selecting features according to the set threshold.

$$w_i = w_i - diff(x_i - H_i)/m + diff(x_i - M_i)/m (5)$$

Where $diff(x_i - H_i)$ and $diff(x_i - M_i)$ are the different on the i-th feature between the X and H(M). m is the number of iterations. According to the above introduction of Relief, we know that the running time of Relief algorithm increases linearly with the increase of the value of m and the feature dimensions. Namely the time complexity is linear. Therefore, Relief is especially suitable for High-dimensional feature space for its high efficiency[36,37]. However, the limitation of only two-class problems solving limits its use. For multi-class data use and regression problems solving, ReliefF is proposed, which is an extension of Relief. It searches for k nearest hits from the same class of X and k nearest misses from each different class. The weight is updated by equation (6).

$$w_i = w_i - \sum_{k=1}^{K} diff\left(x^i - H_k^i\right) \Big/ (m^*K) + \sum_{c \neq c1} \frac{p(c)}{1 - p(c1)} \sum_{k=1}^{K} diff\left(x^i - H_{c,k}^i\right) \Big/ (m^*K) \tag{6}$$

Where w_i is the i-th weight. $p(c)$ is the prior probability of class c. $c1$ is the class of X. k is the number of H and M. It can be seen that the running time of ReliefF increases linearly with the increase of m, feature dimensions and k, namely the time complexity is still linear. Meanwhile, it can handle problems with multiclass datasets and continuous-time regression. Thanks to the applicability of high-dimensional space and the execution efficiency, ReliefF is on the top of list for feature selection in NCBP estimation. This is because that ReliffF can efficiently transfer multi-dimensional features to BP prediction features according to weight attribution. In this situation, ReliefF not only saves computing resources, but also facilitates the portable integration of BP prediction models.

Research on ReliefF for selecting an optimal feature subset to estimate NCBP is not enough. In this study, the topics, "blood pressure" and "ReliefF", were searched in the databases of Web of Science, IEEE Xplore, ScienceDirect, PubMed and Springer. Three related work are found. One of them estimated BP by speech recordings, which was little relevance to the orientation of our study[38]. Another one extracted 24 features from ECG and PPG, 12 optimized features were selected based on ReliefF. After constructing a regularized regression model to estimate BP, the root mean square error (RMSE) of SBP and DBP reached 5.28 mmHg and 5.98 mmHg, respectively[39]. The third one extracted 107 features from PPG and demographic information, then11 and 10 features were selected by ReliefF for SBP and DBP estimation, respectively, then gaussion process regression was used to estimate BP. The RMSE of SBP and DBP reached 6.74 and 3.59 mmHg, respectively[40]. However, all of them ignored a shortcoming of ReliefF that the weight attribution is converted by calculation between the nearest neighbor instances, which does not deal with the redundancy among features. In other words, two feature dimensions with strong correlation will be given approximate weight by ReliefF. Redundancy cannot be eliminated through weights, which reduces the efficiency of model training. Therefore, Pearson correlation analysis was used to remove highly correlated parameters before ReliefF in this study, which could further reduce the dimension of features and improve the efficiency and accuracy of the model.

2.3. BP prediction model based on MPGA-BPN

Following feature screening, the model based on MPGA and BPN is used for NCBP estimation[27]. BPN is a feedforward neural network that can fit complex nonlinear relationships with three layers: input layer, hidden layer and output layer. The information of the selected features is flowed between input and output layer. The flowing is determined by on the weights and thresholds of each neuron of BPN. The initial weights and thresholds are constantly updated during the "forward propagation" and "backward propagation" of network training. When the network meet expectations, the training stops, and the weights and thresholds are fixed.

The initial weight and threshold have a great influence on the training, which will directly affect the performance of the model. The initial weights and thresholds of the untrained BPN are random, so there is a randomness to training results and some results are not ideal. To solve this problem, MPGA developed by Genetic Algorithms is used to determine initial weights and thresholds. It was first proposed by Potts [41] and the basic ideas are natural selection and evolutionary mechanisms. The process of MPGA in this study is as follows:

1. Constructing 5 different populations: Each population has 20 individuals and each individual represents a set of initialized weights and thresholds. We need to ensure that the control parameters of each population are different;
2. Encoding each individual in each population and calculate the fitness of each individual;
3. Using the selection algorithm to select individuals from each population, and generate new individuals according to the crossover and mutation operators;
4. Calculating the fitness of the offspring in step 3;
5. Finding out the individuals with the highest fitness in each population for immigration operations;
6. Forming an elite population with the highest fitness in each population;
7. Determining the individual with the highest fitness in the elite population, and record the algebra that the individual maintains the optimal state in the elite population. When the algebra exceeds the preset algebra, the algorithm terminates, and the individual is decoded into the weights and thresholds for BPN.

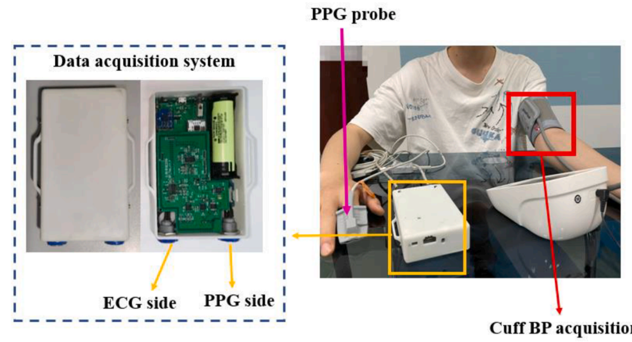

Fig. 2. Schematic diagram of simultaneous acquisition of PPG, ECG and cuff BP.

3. Experiments and results

This section presents the experiments of data collection and the results of feature screening and model prediction. According to the result of feature screening, the number of 3, 5, 7 and all features are selected to constitute four features subsets. each subset serves as input to MPGA-BPN model, and the results of BP prediction are analyzed and compared across subsets. The mean absolute error (MAE), standard deviation (STD) and RMSE are the metrics of accuracy of BP estimation. Then the regression plot is used to further analyze the correlation between predicted and measured BP.

There are two datasets used in this study: CQU_BP dataset and public dataset UCI_BP[42]. CQU_BP is collected from healthy subjects in this study and the collection process will be described in section 3.1.

The public dataset UCI_BP was developed from MIMIC II, and all individuals are patients in ICU. In this study, the data of UCI_BP were segmented with a time window of 20 s. A total of 23,608 sets of data were collected, each consisting of PPG, ECG waveforms and ambulatory invasive BP with a duration of 20 s. The distributions of BP in the UCI_BP are shown in Fig. 3a and Fig. 3b.

3.1. Data acquisition

ECG, PPG signals and cuff BP of CQU_BP dataset were collected synchronously, as shown in Fig. 2. Cuff BP was measured through commercial device (Omron, HEM-7210), whereas the PPG and ECG were recorded by the front-end device independently developed in this study. The front-end measures ECG and PPG signals synchronously and stored them in a file on a TF card. The standard limb lead was used in ECG configuration. The PPG sensor was clamped on the fingertip of right index finger of subjects. The sampling frequency was 500 Hz.

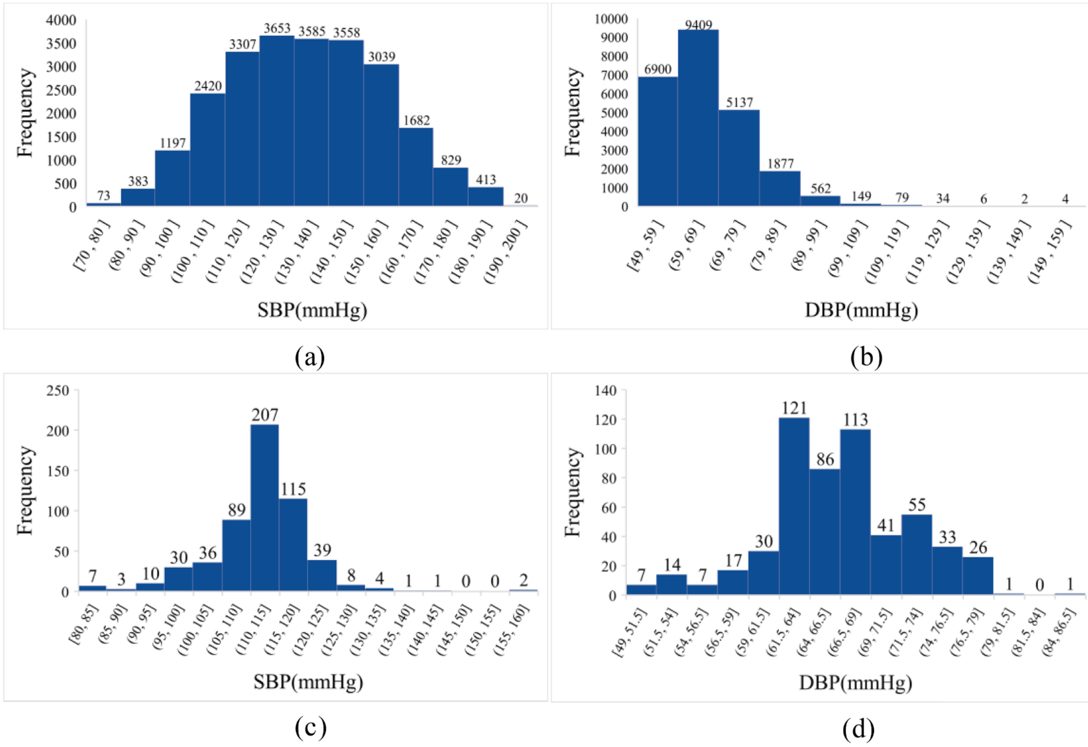

Fig.3. The distributions of BP: SBP of UCI_BP, (b) DBP of UCI_BP (c) SBP of CQU_BP, (d) DBP of CQU_BP.

Table 2
Strongly correlated features and corresponding Pearson correlation coefficients.

	t_{down}	dt_{down}	t_{bb}	t_{fb}	PTT_a	PTT_b	tw2	tw3	tw4	tw5	dS_{bc}	dS_{cb}	A_{fb}	A3	dAA	I_{er}
HR	−0.891	−0.949	−0.963													
t_{down}	1	0.955	0.937													
dt_{down}		1	0.989													
t_{bb}			1													
t_{bf}				−0.864												
PTT_c					0.996	0.991										
PTT_a					1	0.997										
tw1							0.976	0.920	0.841							
tw2							1	0.971	0.906	0.807						
tw3								1	0.968	0.885						
tw4									1	0.956						
S_{bc}											0.924	−0.828				
dS_{cb}											−0.824	1				
S_{cb}												0.828			−0.828	
A_{bf}														0.813		
A4				0.832									0.936			
dDA															0.919	
I_{fr}																0.822

A total of 25 healthy subjects (12 females and 13 males) aged 18–27 years were included in CQU_BP. All procedures were performed with the written informed consent. During the experiment, the subjects sat in the armchair, PPG sensor was clipped to the right index finger of the subjects, and the ECG electrodes were attached to human following three-lead cables (LA, RA, LL). The cuff was worn on the left upper arm to measure BP. PPG and ECG signals were recorded when the subjects have rested for 5 min. After the waveform of signals became stable, BP was measured every 10 min, and each BP measurement was completed within 1 min. The duration of each signals acquisition was about 20 min. Following the signal acquisition, the PPG and ECG and their BP reference were segmented into 20 s window. Removing the waveforms without reference BP, we collected a total of 552 sets of data. Each set included the PPG and ECG waveforms with a duration of 20 s of individuals, and corresponding SBP and DBP. The distributions of BP from CQU_BP are shown in Fig. 3c and Fig. 3d.

3.2. Final determination of model parameters

Based on the methods described in Section 2, Pearson correlation analysis was performed on the extracted 34 features. The strongly correlated features and corresponding Pearson correlation coefficients are shown in Table 2 (The complete results of Pearson correlation analysis are in supplementary material). Table 2 shows that t_{down}, dt_{down}, t_{bb} and HR are highly correlated; t_{fb}, A4 and t_{bf} are highly correlated; PTT_a, PTT_b and PTT_c are highly correlated; tw1, tw2, tw3, tw4 and tw5 are highly correlated; S_{bc}, dS_{bc}, dS_{cb} and S_{cb} are highly correlated; A_{fb} and A4 are highly correlated; A_{bf} and A3 are highly correlated; dAA and dDA are highly correlated; I_{er} and I_{fr} are highly correlated. In this study, A total of 14 parameters, dt_{down}, t_{bb}, t_{bf}, PTT_a,

Table 3
Weight ranking of feature parameters.

Rank	SBP	DBP
1	t_{ae}	t_{ae}
2	K	K
3	S_{bc}	t_{fb}
4	A3	A3
5	t_{down}	t_{down}
6	t_{fb}	A1
7	HR	t_{up}
8	t_{up}	I_{er}
9	tw1	A4
10	S_{cf}	HR

PTT_b, tw2, tw3, tw4, dS_{bc}, dS_{cb}, A_{fb}, A_{bf}, dDA and I_{fr} were excluded. The basic principle of exclusion is to exclude the features that are not easily extracted among highly correlated parameters, which is beneficial to improve the robustness of the whole system in the later engineering implementation research.

RA solved the problem that ReliefF could not handle the redundancy between features, which helped to improve the efficiency of model training. In addition, when using ReliefF for features selection, similar parameters were excluded, so the features of the same dimension would contain more information of signals, which was beneficial to the estimation of NCBP.

Following the RA, ReliefF was further used to process the remaining features to obtain the weight of each feature. The ranking of the top ten features based on weights is given in Table 3.

3.3. Accuracy of models

Two datasets in this study are from healthy individuals and patients in the ICU, so the models will be trained and tested separately for the CQU_BP and UCI_BP datasets to further observe the feasibility of BP estimation with fewer parameters. We select the top three, five, seven and all parameters after RA-ReliefF to form the three-parameters, five-parameters, seven-parameters and full-parameters datasets for the CQU_BP dataset and the UCI_BP dataset, respectively. After the MPGA-BPN models were trained, the SBP and DBP performance of the model on the two datasets are shown in Fig. 4, Fig. 5, Fig. 6 and Fig. 7.

Fig. 4 shows the comparison between the predicted SBP and cuff SBP in CQU_BP dataset. Fig. 4(a) shows performance on SBP estimation of three-parameters set, and it can be found that the overall trend on the predicted values can be consistent with the cuff BP, but three-parameters set is not sufficient for SBP estimation when BP values are relatively extreme. Fig. 4(b), (c), (d) shows the comparison of predicted SBP and cuff SBP for five, seven and all-parameters sets, respectively. It can be seen that the performance of 5, 7 and all parameters is in close agreement. We therefore believe that the performance of SBP estimation with all-parameters can be achieved with five- parameters set on CQU_BP dataset.

Fig. 5 shows the comparison between the predicted DBP and cuff DBP in CQU_BP dataset. Fig. 5 (a), (b), (c), and (d) show performance on DBP estimation of three, five, seven and all-parameters sets. It can be

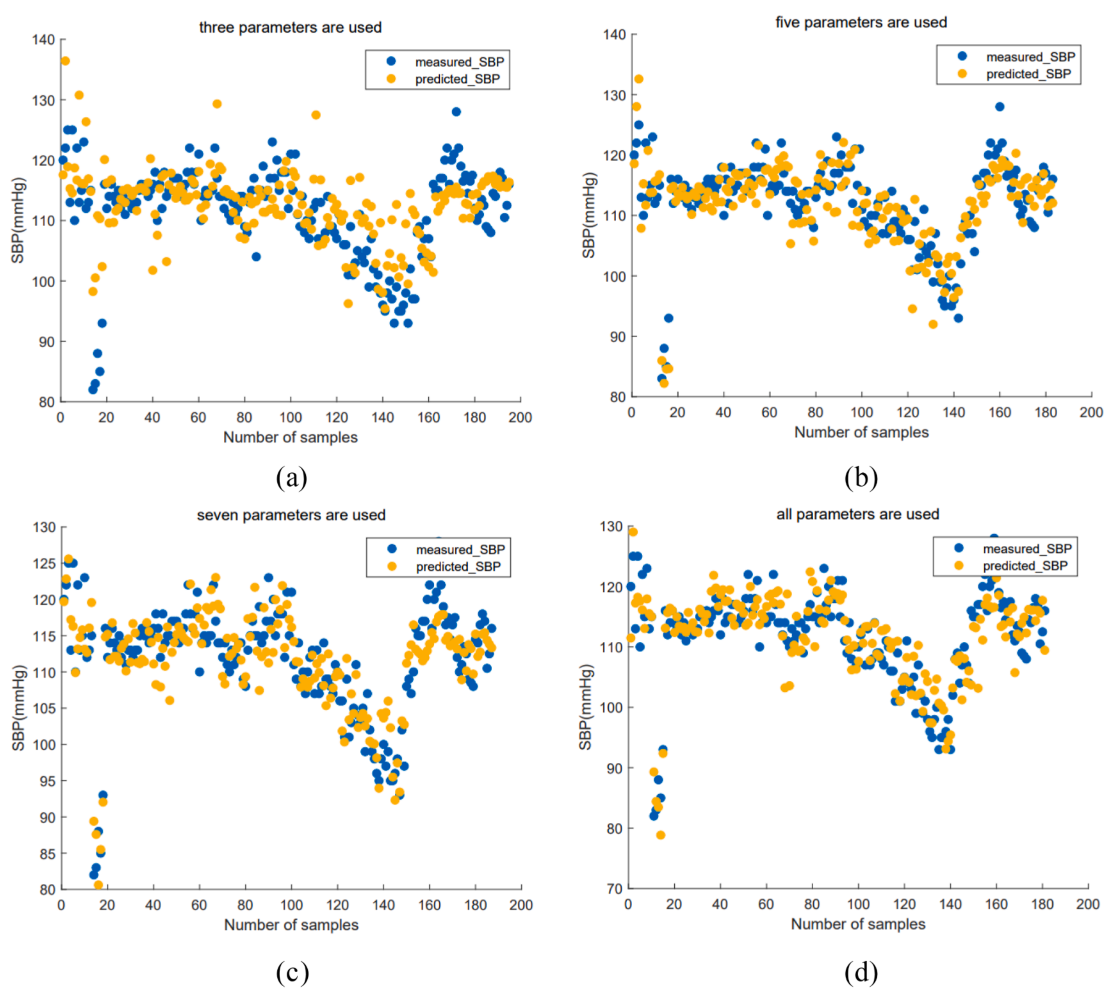

Fig. 4. Comparison of predicted and measured values of SBP for different parameters sets in CQU_BP. (a) three-parameters. (b) five-parameters. (c) seven-parameters. (d) full-parameters.

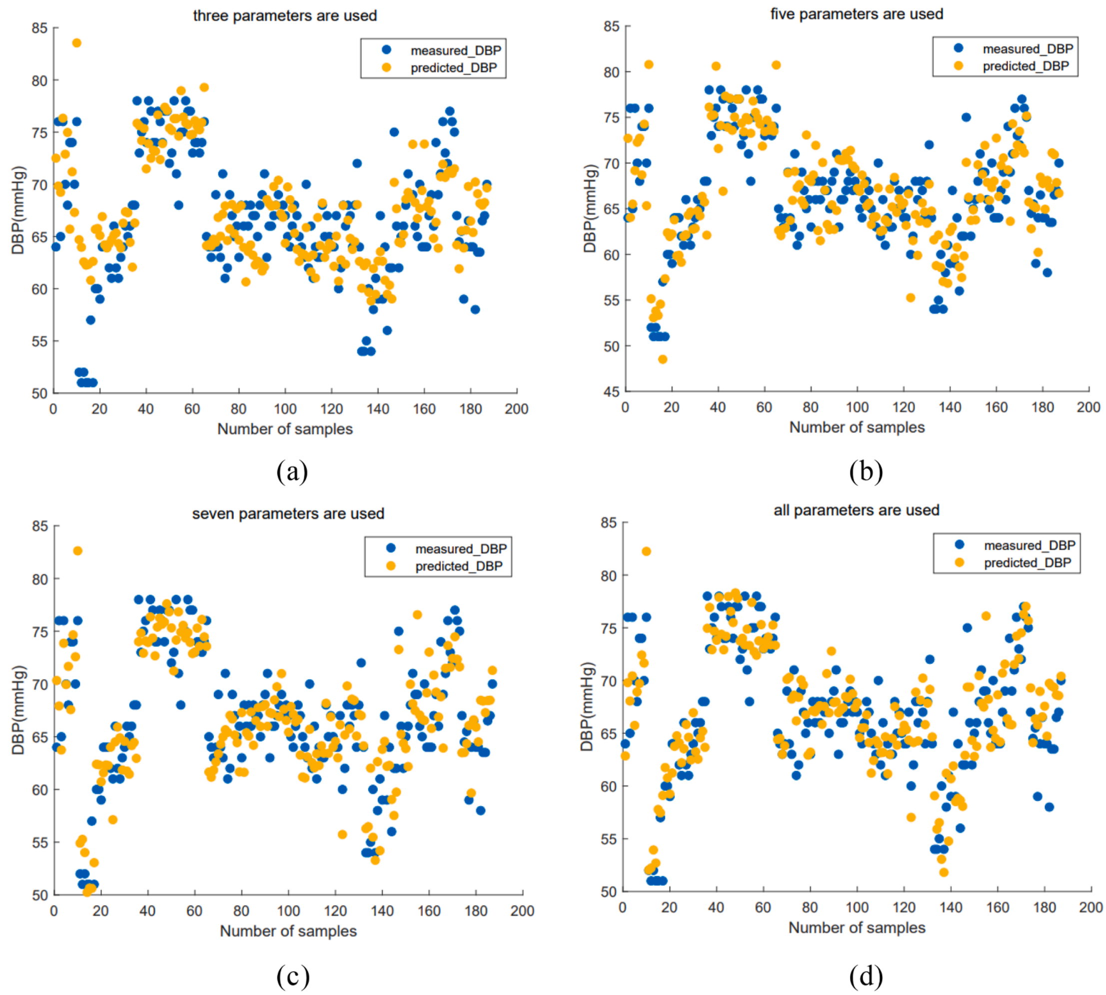

Fig. 5. Comparison of predicted and measured values of DBP for different parameters sets in CQU_BP. (a) three-parameters. (b) five-parameters. (c) seven-parameters. (d) full-parameters.

seen that they are in close agreement. We therefore believe that the performance of DBP estimation with all-parameters can be achieved with three-parameters set on CQU_BP dataset.

Fig. 6 and Fig. 7 show the comparison between the predicted values and invasive measured values of SBP and DBP in UCI_BP dataset. Figures (a), (b), (c), and (d) show performance on BP estimation of three, five, seven and all-parameters sets. It can be seen that the performance of 5, 7 and all parameters do not differ much in both SBP and DBP estimation. By comparison, the consistency between predicted and measured BP of three-parameters set is worse than the other feature sets. We therefore believe that the performance of SBP and DBP estimation with all-parameters can be achieved with five-parameters set on UCI_BP dataset.

Table 4 presents the performance of the MPGA-BPN model on CQU_BP and UCI_BP dataset. MAE (Equation (7)), STD (Equation (8)) and RMSE (Equation (9)) are the metrics. Among them, STD shows the standard deviation of the difference between the predicted and true values. y_i is the BP measured by the sphygmomanometer or measured invasively, $\widehat{y}_i$ is the BP predicted by the model, $\overline{y_i - \widehat{y}_i}$ is s the mean of the difference between the predicted and true values, n is the number of samples.

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \widehat{y}_i| \tag{7}$$

$$\mathrm{STD} = \sqrt{\left(\sum\nolimits_{i=1}^{n}(y_i - \widehat{y}_i - \overline{y_i - \widehat{y}_i})^2\right)\Big/(n-1)} \tag{8}$$

$$\mathrm{RMSE} = \sqrt{\left(\sum\nolimits_{i=1}^{n}(y_i - \widehat{y}_i)^2\right)\Big/(n-1)} \tag{9}$$

Table 4 shows that all feature sets in CQU_BP dataset are up to AAMI standards when estimating SBP and DBP (MAE<=5mmHg, STD<=8mmHg). In terms of estimating SBP, there is no significant difference in MAE and STD between the five-parameters, seven-parameters and full-parameters feature sets. For DBP estimating, the performance of three-parameters, five-parameters, seven-parameters and full-parameters sets are all ideal and the metrics are similar. In summary, in order to minimize the dimension of parameters, reduce the redundancy of features and the complexity of the model, and ensure the accuracy of BP estimation, the five-parameters feature set is superior for SBP estimation and three-parameters feature set is superior for DBP estimation in CQU_BP dataset. This result is consistent with that derived from the analysis in Fig. 4 and Fig. 5 above.

In UCI_BP dataset, the feature sets are not up to AAMI criteria in estimating SBP and DBP. However, it can be stated that the five-parameter, seven-parameter, and full-parameter feature sets did not differ significantly in MAE and STD. Namely, the performance of SBP and DBP estimation with all-parameters can be achieved with five-

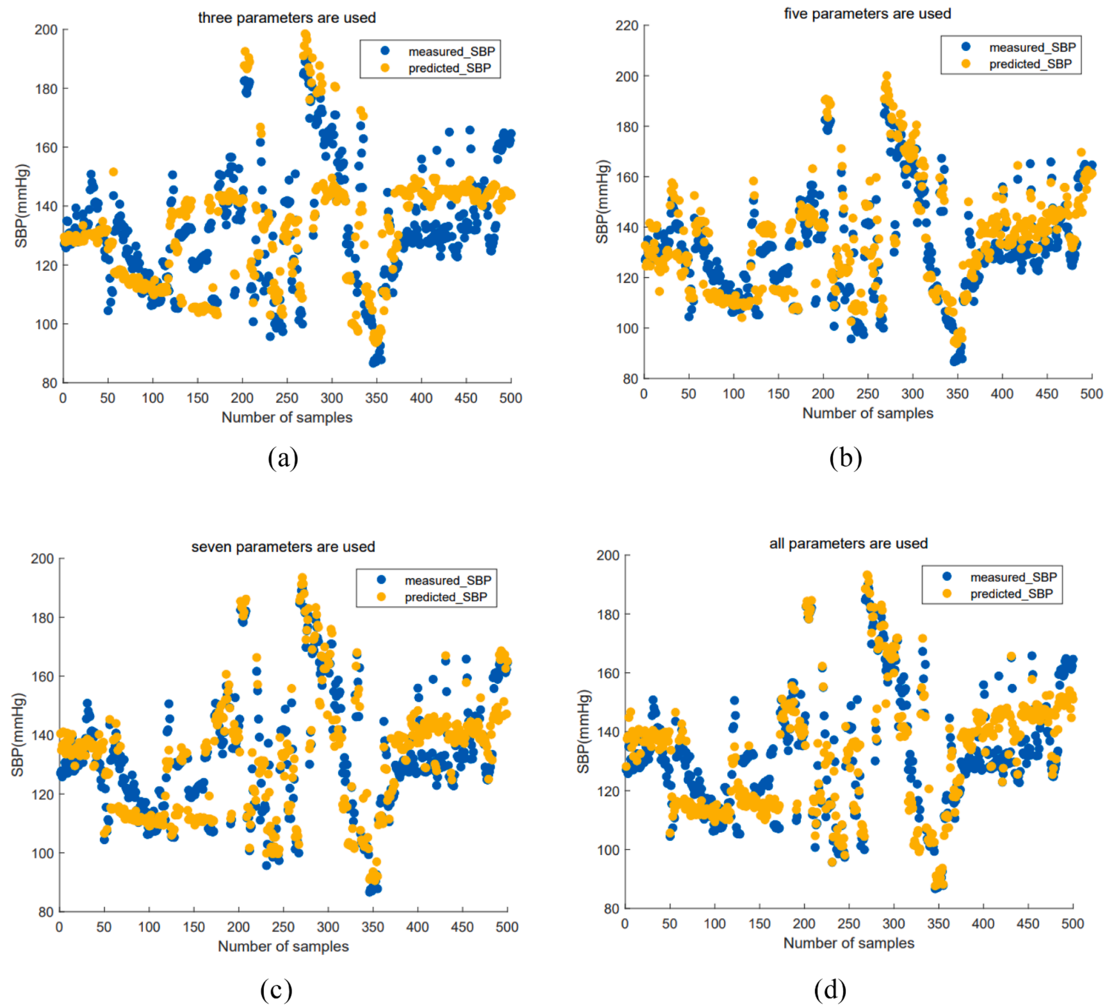

Fig. 6. Comparison of predicted and measured values of SBP for different parameters sets in UCI_BP. (a) three-parameters. (b) five-parameters. (c) seven-parameters. (d) full-parameters.

parameters set, which indicates the validity of feature selection in this study.

Summing up, for CQU_BP dataset, five-parameters and three-parameters sets are sufficient for SBP and DBP estimation, respectively. For UCI_BP dataset, both SBP and DBP estimation require five-parameter sets. Comprehensive consideration, this study concluded that a subset of five-parameters could be used for BP estimation instead of full-parameters set without reducing the accuracy.

Fig. 8 presents a regression diagram of the measured and predicted BP of the five-parameters for CQU_BP and UCI_BP datasets, where the abscissa represents the target BP (measured BP), and the ordinate represents the predicted BP. Fig. 6(a) and (b) are the SBP and DBP of CQU_BP; (c) and (d) are the SBP and DBP of UCI_BP, respectively. Each diagram includes the degree of correlation between the target and predicted values of the training set, validation set, test set, and all data sets. It can be seen that there is a strong correlation between the target and predicted value for both CQU_BP and UCI_BP datasets. The correlation coefficient R is greater than 0.8 in CQU_BP dataset and close to 0.7 in UCI_BP, indicating that the validity of the five-parameter feature set estimation of blood pressure in this study.

4. Discussion

In this study, PPG and ECG were collected through experimental and public dataset UCI_BP, and 34 parameters were extracted. 14 parameters with high degree of correlation were eliminated by RA. In order to estimate BP with fewer parameters and improve the stability and accuracy of the system, the ReliefF algorithm was further used to process the remaining parameters for corresponding weight. Then, the most important 3, 5, 7 and all parameters were selected based on the weight value for BP estimation. The experimental result in Table 4 shows that on the premise of ensuring the accuracy of BP estimation, the five parameters screened by the RA and ReliefF algorithm are sufficient for the estimation of SBP and DBP in both CQU_BP and UCI_BP datasets. The regression plot shown in Fig. 6 demonstrates the feasibility of this study to achieve BP estimation with high accuracy and low error by five parameters.

In our previous study[27], 14 features were selected from 24 features by MIV combined with the BPN, and the MPGA-BPN model clusters were constructed to estimate BP. The MAE $\pm$ STD of predicted SBP and DBP were 2.59 $\pm$ 3.41 mmHg and 2.69 $\pm$ 3.31 mmHg, respectively, which were similar to the accuracy of the five-parameter BP prediction in this study, indicating the effectiveness of the feature screening in this study. Moreover, the complexity of parameter selection and model in previous studies were both greater than those in current study, so the current study is more favorable for the integration of NCBP on portable devices.

In order to achieve NCBP estimation, immense efforts have been made by researchers in physiological signals. The extraction and selection of signal features are crucial. Regardless of extracting features automatically or manually [43,44], the number of parameters are needed to be further reduced to decrease the complexity of the model and improve the stability of the BP estimation system, so NCBP

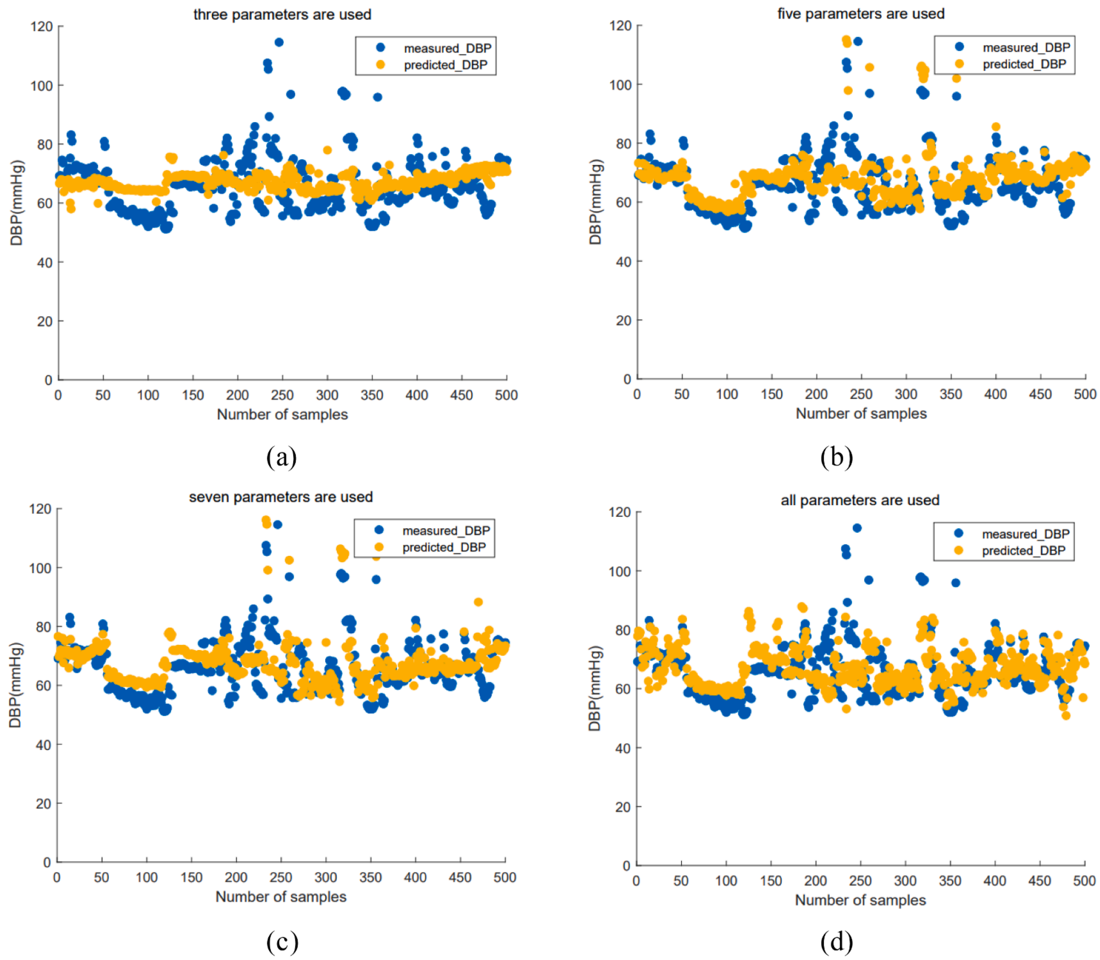

Fig. 7. Comparison of predicted and measured values of DBP for different parameters sets in UCI_BP. (a) three-parameters. (b) five-parameters. (c) seven-parameters. (d) full-parameters.

Table 4
Performance comparison of feature sets in predicting BP.

datasets	BP	Feature set	MAE	RMSE	STD
CQU_BP	SBP	Three features	4.68	6.42	6.32
		Five features	3.18	3.98	3.97
		Seven features	3.11	3.94	3.92
		All features	3.07	3.97	3.97
	DBP	Three features	2.87	3.73	3.78
		Five features	2.48	3.26	3.25
		Seven features	2.55	3.22	3.21
		All features	2.37	3.12	3.07
UCI_BP	SBP	Three features	9.83	11.36	11.18
		Five features	7.01	7.70	6.92
		Seven features	6.51	8.23	8.19
		All features	6.50	8.21	8.16
	DBP	Three features	7.95	12.70	12.70
		Five features	6.18	9.19	8.86
		Seven features	6.11	8.92	8.59
		All features	6.39	8.93	8.93

estimation can be freed from laboratory stage and the process of productization and clinicalization can be accelerated.

On the premise of ensuring the accuracy of BP estimation, this study uses 5 parameters to estimate SBP and DBP, which reduced the redundancy of features and the influencing factors of BP estimation. simultaneously, the points that need to be located in the feature point localization can be reduced in future studies, so the stability of the positioning algorithm in different situations is improved, which further improves the stability and accuracy of the BP estimation. Moreover, the time complexity of the ReliefF for parameter dimensions reduction in this study is linear, which is so efficient that will not increase the burden of the model.

However, ReliefF ignores the relevance between dimensions. Dimensions with high correlation will get approximate weights. Therefore, compared with the BP estimation studies that only use ReliefF for feature selection, this study has considered the inherent defect of ReliefF, RA is carried out before ReliefF, so the selection of features with the same dimensions contained more physiological information. It is more beneficial to use less parameter information and improve the accuracy of BP estimation. In this study, five parameters were used to estimate BP, while the number of parameters selected by other ReliefF related studies was about 10.

Overall, this study achieves the same precision as all-parameters NCBP estimation via the MPGA-BPN model by 5 parameters. The parameter dimensions are reduced, so the complexity of the model is reduced. In the following work, because the reduction of parameters reduces the influencing factors of BP prediction, we can effectively reduce the positioning of unnecessary points among the feature point localization. In this way, the operation efficiency, stability and accuracy of the BP estimation model are improved, which is further beneficial for the clinical application of the NCBP.

5. Conclusion

In this study, we achieved BP estimation with the same accuracy as

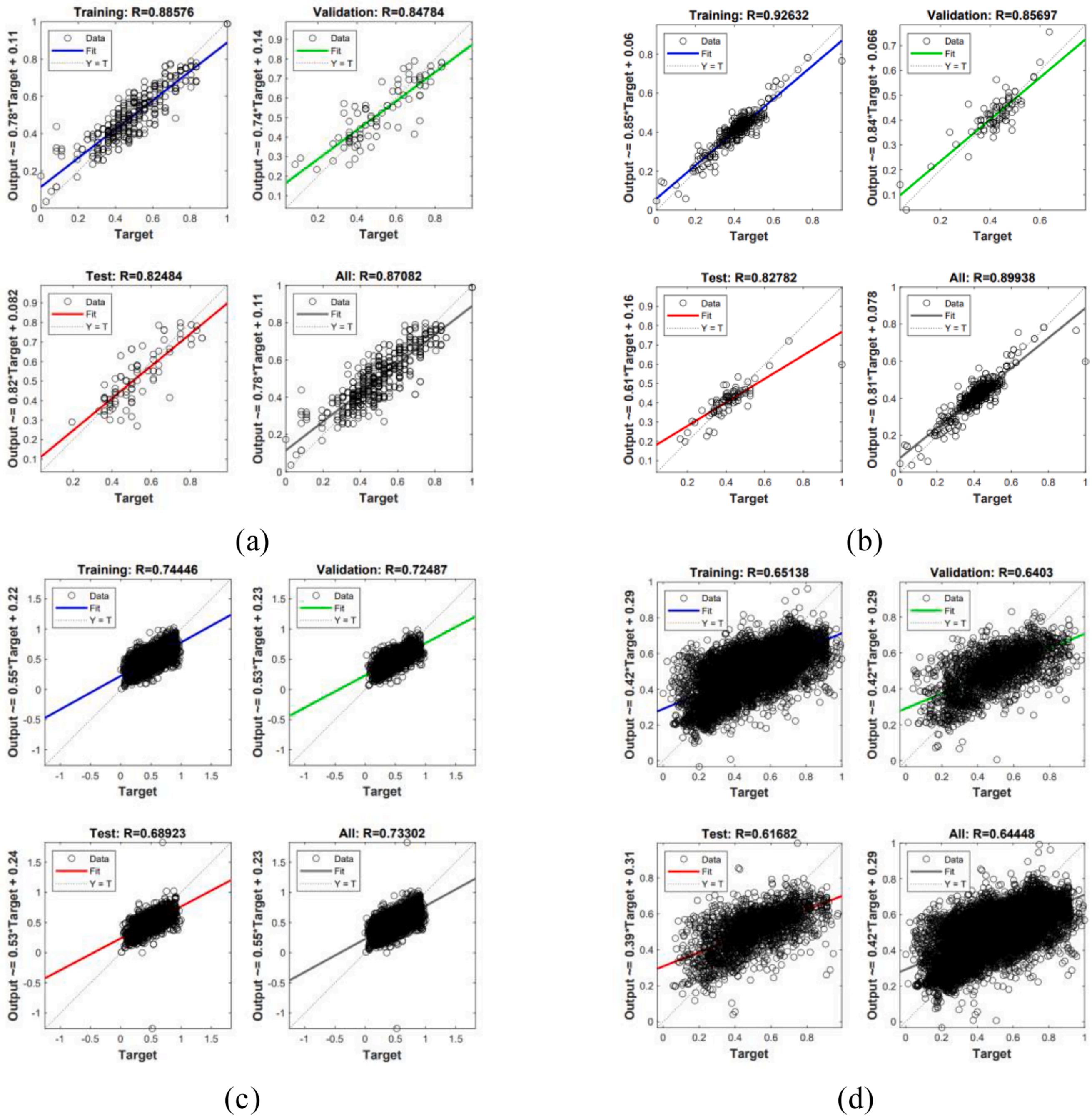

Fig. 8. Regression plot of measured and predicted BP of five parameters (a) SBP of CQU_BP. (b) DBP of CQU_BP. (c) SBP of UCI_BP. (d) DBP of UCI_BP.

the full parameters by fewer parameters. In the data collection process, we fully consider the influence of different populations, and collect data from healthy people to form the CQU_BP dataset and the public dataset of non-healthy people, UCI_BP. In the feature selection process, the shortcoming that ReliefF cannot handle the redundancy among features is fully considered, and the extracted features are selected by combining RA and ReliefF, which effectively excludes features with high correlation degree, so that the same number of feature dimensions contains more physiological information. In terms of modeling, this study estimated BP based on the developed MPGA-BPN model, and our results show that five-parameters set reaches the same accuracy of the full-parameters BP estimation. Thus, our method effectively selects the signal information required for BP estimation, significantly reduces the number of features without decreasing the accuracy of BP estimation, and effectively reduces the redundancy of features and the complexity of the model. In the subsequent work, the reduction of parameters leads to fewer influencing factors in BP prediction, and the feature point localization process can effectively reduce the localization of non-essential feature points, which improves the operational efficiency, stability and accuracy of models and is more conducive to the clinical application of BP estimation methods.

Ethical approval.

Ethical approval was given by Chongqing University's College of Biological Engineering. The participants were volunteers. The volunteers were informed of the specific process and risk of the experiment beforehand, and the collection of data was carried out with the volunteer's consent. There was no conflict of interest in this research.

CRediT authorship contribution statement

Lanli Zhang: Conceptualization, Methodology, Validation, Writing – original draft. **Zhong Ji:** Supervision, Resources, Writing – review & editing, Funding acquisition. **Fan Yang:** Visualization, Investigation. **Gang Chen:** Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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References

[1] Health World Organization, "World health statistics 2021: monitoring health for the sdgs," Licence, (2021).
[2] L.C.C. Brant, L.G. Passaglia, The burden of resistant hypertension across the world, Curr. Hypertens. Rep. 24 (2022) 55–66.
[3] Z. Wang, A. Du, Systematic analysis of the global, regional and national burden of cardiovascular diseases from 1990 to 2017, Journal of Epidemiology and Global Health 12 (2022) 92–103.
[4] J. Fortin, W. Marte, Continuous non-invasive blood pressure monitoring using concentrically interlocking control loops, Computers In Biology And Medicine 36 (2006) 941–957.
[5] A. Chandrasekhar, M. Yavarimanesh, Ppg sensor contact pressure should be taken into account for cuff-less blood pressure measurement, IEEE Trans. Biomed. Eng. 67 (2020) 3134–3140.
[6] R. Mukkamala, J. Hahn, Toward ubiquitous blood pressure monitoring via pulse transit time: theory and practice, IEEE Trans. Biomed. Eng. 62 (2015) 1879–1901.
[7] M. Elgendi, On the analysis of fingertip photoplethysmogram signals, Curr Cardiol Rev 8 (2012) 14–25.
[8] J.C. Bramwell, The velocity of the pulse wave in man, Proceedings Of The Royal Society Of London Series B-Containing Papers Of A Biological Character 93 (1922) 298–306.
[9] D.J. Hughes, C.F. Babbs, Measurements of youngs modulus of elasticity of the canine aorta with ultrasound, Ultrason. Imaging 1 (1979) 356–367.
[10] J. Ji and M. Dong, "Noninvasive cuffless blood pressure estimation with dendritic neural regression," IEEE TRANSACTIONS ON CYBERNETICS, pp. 1-13, (2022).
[11] H.Q. Wang, Z.G. Wang, A novel approach to estimate blood pressure of blood loss continuously based on stacked auto-encoder neural networks, Biomedical Signal Processing And Control 69 (2021).
[12] M. Das, T. Choudhary, Noninvasive accelerometric approach for cuffless continuous blood pressure measurement, IEEE Transactions On Instrumentation And Measurement 70 (2021).
[13] Y.J. Ma, J. Choi, Relation between blood pressure and pulse wave velocity for human arteries, PNAS 115 (2018) 11144–11149.
[14] V. Figini, S. Galici, Improving cuff-less continuous blood pressure estimation with linear regression analysis, Electronics 11 (2022).
[15] H. Shin, A novel method for non-invasive blood pressure estimation based on continuous pulse transit time: an observational study, Psychophysiology 60 (2023).
[16] C. Fischer, T. Penzel, Continuous non-invasive determination of nocturnal blood pressure variation using photoplethysmographic pulse wave signals: comparison of pulse propagation time, pulse transit time and rr-interval, Physiol. Meas. 40 (2019).
[17] E. Monte-Moreno, Non-invasive estimate of blood glucose and blood pressure from a photoplethysmograph by means of machine learning techniques, Artif. Intell. Med. 53 (2011) 127–138.
[18] X. Chen, S. Yu, Machine learning method for continuous noninvasive blood pressure detection based on random forest, IEEE Access 9 (2021) 34112–34118.
[19] K. Huang, F. Tan, A highly sensitive pressure-sensing array for blood pressure estimation assisted by machine-learning techniques, Sensors 19 (2019).
[20] S. Baek, J. Jang, End-to-end blood pressure prediction via fully convolutional networks, IEEE Access 7 (2019) 185458–185468.
[21] C. Yen, S. Chang, Development of a continuous blood pressure measurement and cardiovascular multi-indicator platform for asian populations by using a back propagation neural network and dual photoplethysmography sensor signal acquisition technology, J. Nanomater. 2021 (2021).
[22] J. Park, D. Kang, Estimation of arterial blood pressure based on artificial intelligence using single earlobe photoplethysmography during cardiopulmonary resuscitation, J. Med. Syst. 44 (2019).
[23] S. Masum and J. P. Chiverton, "Investigation of machine learning techniques in forecasting of blood pressure time series data," in *ARTIFICIAL INTELLIGENCE XXXVI*. vol. 11927, M. Bramer and M. Petridis, Eds. 39th SGAI International Conference on Techniques and Applications of Artificial Intelligence (AI), 2019, pp. 269-282.
[24] A. Argha, B.G. Celler, Blood pressure estimation from time-domain features of oscillometric waveforms using long short-term memory recurrent neural networks, IEEE Transactions On Instrumentation And Measurement 69 (2020) 3614–3622.
[25] H. Mou, J. Yu, Transfer learning with dwt based clustering for blood pressure estimation of multiple patients, Journal Of Computational Science 64 (2022).
[26] M. Kachuee, M.M. Kiani, Cuffless blood pressure estimation algorithms for continuous health-care monitoring, IEEE Trans. Biomed. Eng. 64 (2017) 859–869.
[27] H. Wu, Z. Ji, Non-invasive continuous blood-pressure monitoring models based on photoplethysmography and electrocardiography, Sensors 19 (2019) 5543.
[28] C. El-Hajj, P.A. Kyriacou, Deep learning models for cuffless blood pressure monitoring from ppg signals using attention mechanism, Biomedical Signal Processing And Control 65 (2021), 102301.
[29] Y.C. Hsu, Y.H. Li, Generalized deep neural network model for cuffless blood pressure estimation with photoplethysmogram signal only, Sensors 20 (2020).
[30] J. Esmaelpoor, M.H. Moradi, Cuffless blood pressure estimation methods: physiological model parameters versus machine-learned features, Physiol. Meas. 42 (2021).
[31] S.S. Mousavi, M. Firouzmand, Blood pressure estimation from appropriate and inappropriate ppg signals using a whole-based method, Biomedical Signal Processing And Control 47 (2019) 196–206.
[32] M. Elgendi and Y. Liang, "Toward generating more diagnostic features from photoplethysmogram waveforms.," Diseases (Basel, Switzerland), 6, (2018).
[33] W.H. Lin, X. Li, Investigating the physiological mechanisms of the photoplethysmogram features for blood pressure estimation, Physiol. Meas. 41 (2020), 044003.
[34] R. J. L and N. W. A,, Thirteen ways to look at the correlation coefficient, Am. Stat. 42 (1988).
[35] M. Robnik-Sikonja, I. Kononenko, Theoretical and empirical analysis of relieff and rrelieff, Mach. Learn. 53 (2003) 23–69.
[36] O. Reyes, C. Morell, Scalable extensions of the relieff algorithm for weighting and selecting features on the multi-label learning context, Neurocomputing 161 (2015) 168–182.
[37] X. Cui, Y. Li, A novel filter feature selection algorithm based on relief, Appl. Intell. 52 (2022) 5063–5081.
[38] H. Ankishan, Blood pressure prediction from speech recordings, Biomedical Signal Processing And Control 58 (2020).
[39] M.M.R. Khan Mamun, A.T. Alouani, Cuffless blood pressure measurement using linear and nonlinear optimized feature selection, Diagnostics 12 (2022) 408.
[40] M.H. Chowdhury, M. Shuzan, Estimating blood pressure from the photoplethysmogram signal and demographic features using machine learning techniques, Sensors 20 (2020).
[41] J.C. Potts, T.D. Giddens, The development and evaluation of an improved genetic algorithm-based on migration and artificial selection, IEEE Transactions On Systems Man And Cybernetics 24 (1994) 73–86.
[42] M. Kachuee and M. M. Kiani, "Cuff-less high-accuracy calibration-free blood pressure estimation using pulse transit time," in *2015 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS)* IEEE International Symposium on Circuits and Systems (ISCAS), 2015, pp. 1006-1009.
[43] X. Chen, H. Chang, Prediction of continuous blood pressure using multiple gated recurrent unit embedded in senet, Journal Of Advanced Computational Intelligence And Intelligent Informatics 26 (2022) 256–263.
[44] T. Treebupachatsakul, A. Boosamalee, Cuff-less blood pressure prediction from ecg and ppg signals using fourier transformation and amplitude randomization preprocessing for context aggregation network training, BIOSENSORS-BASEL 12 (2022).